

VALENTENO LENORA

AI/ML ENGINEER | FULLSTACK DEVELOPER | DATA ENGINEER

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PROFILE SUMMARY

Currently developing AI systems at AiRentoSoft and WaveSkill, with expertise spanning Agentic Systems, Deep learning, Machine learning and scalable infrastructure. Dual-degree background in Computer Science (First Class Honours) and Information Systems provides both technical depth in AI/ML and Business systems perspective. Published a research paper and Open Source package for community. Passionate about transforming complex AI innovations into deployed solutions that deliver measurable real-world impact.

WORK EXPERIENCE

AI/ML Engineer - Freelance - WaveSkill Pvt. Ltd. 2025 (December) - Present

- Designed and Engineered the AI-powered financial document extraction pipeline with computer vision models and helper agents to extract financial data from company statements.
- Developed the fundamental analysis engine for the Stock Market API to give users stats and AI-powered insights on companies regarding their financial positions.
- Developed an Agentic Shopping assistant for Baby Care products e-commerce website (TOGO).
- Developed a TTS dataset and trained a VITS Model for Sinhala.

AI/ML Engineer - Intern - AiRentoSoft Pvt. Ltd. 2026 (March) - 2026 (September) | 6 months

- Developed a vehicle damage detection system with DL models, API and Mobile app.
- Collaborated in further developing the AiRentoSoft System which includes the Web app, Mobile app, Reservations Plugin and AI Caller Agent.
- Assisted in business leads extraction for potential clients for improving business opportunities.
- Collaborated with the developers, business support team and QA engineers via Jira with scheduled sprints and meetings.

AI/ML Engineer - Intern - Residue Solutions Pvt. Ltd. 2025 (September) - 2026 (February) | 6 months

- Engineered an AI-powered shopping assistant (FashionHub.ai) for customers to get assistance on product search, Virtual Try-On, Cart management, Payment assistance, QnA and Size matching.
- Leader of the AI/ML Intern team of five, providing technical guidance on project architecture, code reviews, and implementation strategies and regular meetings to ensure milestones were met.
- Conducted research and integration by collaborating and leading the team members with LLM orchestration techniques, reasoning and multi-model pipelines that enabled complex product recommendations and Virtual Try-On features.

System Support Officer - IT - Mercantile Investments FPLC 2022 | 3 months

- Assisted in hardware, network maintenance and repairs, significantly reducing issues for users through proactive diagnostics and component-level troubleshooting. Resolved technical queries in Help Desk support, providing rapid software, OS, and peripheral solutions for diverse end-user requirements.

EDUCATION

BSc (Hons) Computer Science (First Class Honours) 2022 - 2025

University of Plymouth, UK (NSBM Green University)

BComp (Hons) Information Systems (4th Year) 2023 - Present

University of Sri Jayewardenepura

Dip. in Technology 2019

Esoft Metro Campus

A/L - Math Stream (A1, C2 (O/L: A7, B2)) 2008 - 2021

St. Peter's College, Colombo - 4

PROJECTS

Work Projects

Vehicle Damage Detection System

YOLO, PyTorch, FastAPI, React, HuggingFace, WandB, Kaggle, MLOps, Roboflow

Developed a Vehicle Damage Detection system consisting of a combination of YOLO models with a FastAPI wrapper hosted in Huggingface spaces. Pre-trained the models on Image datasets, versioned and deployed using MLOps tools. Also further developed the live mobile app to support new features which is in Playstore and Appstore.

AiRentoSoft System

React, Node.js, .NET, Retel.ai, OCR, Azure DevOps

Collaborated in Developing and Implementing new features including AI Integrations. Worked on the React Web application, Reservations Plugin, .NET API, and developed the 2.0 version of the Mobile app with similar features to the main system and released documentation in AZURE DevOps. Added AI Integrated features like document scanning, card scanning using OCR models in both web and mobile apps. Also managed and worked on the AI Caller agent in Retel.ai for client requirements.

Business Leads Extraction

Playwright, Selenium, Python, ETL

Implemented a business leads extraction pipeline for extracting potential clients in USA based car rental businesses using Automation scripts and web scraping and using ETL pipelines to process data into understandable formats for business support division to use.

Sinhala TTS Model

Coqui VITS, PyTorch, Kaggle

Developed a research purpose private TTS model with support for Sinhala language. Created a private dataset from scratch for VITS format to train the Speech Model. Trained on Kaggle dual T4 GPUs and gained research level quality output for product pitching and further development with funds.

TOGO - AI Shopping Assistant

FastAPI, Postgres, CrewAI, LangGraph, Langfuse

Developed an Agentic Shopping assistant for e-commerce website for baby care products. Implemented features in Product search, QnA using store policies and customer support. This is live on the TOGO website.

FashionHub.ai

Next.js, FastAPI, Weaviate, Ollama, LangGraph, LangChain, Kafka, Groq, Gemini, MongoDB, Google Vertex AI

Developed an intelligent AI shopping assistant that revolutionises the retail experience by utilising a ReAct capable Agent with DAG-based task execution. The platform features a high-throughput MongoDB to Weaviate Kafka pipeline for real-time data synchronization. Built on a scalable FastAPI microservices infrastructure, the system integrates a modular multi-LLM factory (Gemini, Llama) and Google Vertex AI for generative Virtual Try-On capabilities, delivering a seamless and context-aware customer assistance. Used LangChain & LangGraph for LLM orchestration and agent state management.

Stock Market API

FastAPI, LlamaIndex, OpenCV, Groq, MongoDB, LLMs, OCR

Designed and engineered a scalable financial intelligence API using FastAPI for Fundamental analysis of Colombo Stock Exchange (CSE) filings. Built the analysis engine that can calculate core financial ratios and growth metrics, integrating AI-driven insights to provide comprehensive qualitative and quantitative evaluations of corporate performance. Additionally, developed an extraction engine that utilizes hybrid OCR and semantic mapping with Large Language Models (LLMs) to transform unstructured quarterly statements into standardized datasets.

Personal Projects

Accordo.ai - Music Analysis Platform

TensorFlow, RNN, Bi-LSTM, Flutter, FastAPI, Celery, Redis, MongoDB, TensorFlow Serving, GCP, MinIO S3

Developed an intelligent system that performs musical analysis on audio for musicians. The project explored advanced deep learning techniques, including Bi-LSTM networks and time-frequency feature extraction, trained on real-world music datasets and achieved +80% accuracy. Focused on research and development of robust music information retrieval (MIR) model, with a practical Implementation using FastAPI, MongoDB, and TensorFlow Serving with a Flutter-based mobile UI for real-time analysis and playback integration. Deployed using a CI/CD pipeline to GCP.

Univize - University Social App

Django, FastAPI, Next.js, Neo4j, PostgreSQL (Supabase), MongoDB, Redis, Supabase, Azure, GitHub Actions, Azure Bicep

Developed an Social application for university students with features including, medial sharing, direct messaging and forum based discussions. Led the development team in system design to development, testing and deployment. Used Azure as the hosting platform with services like container registry, secret management and containerized apps. Developed the main backend in Django and FastAPI with PostgreSQL (Supabase), MongoDB, Neo4J, Redis and Supabase. Built the CI/CD pipeline with GitHub Actions and Azure Bicep scripts for infrastructure provisioning and Integration. This was done as the Group project for USJP.

EduAgent - AI-powered Research Assistant

CrewAI, LangChain, Ollama, Mistral LLM, ChromaDB, HuggingFace

Developed a Research assistant leveraging LangChain, CrewAI, and Mistral LLM (via Ollama) for research paper analysis and academic Q&A. Implemented document ingestion, summarization, topic extraction, and semantic retrieval using HuggingFace sentence-transformer embeddings with ChromaDB. Designed modular agent architecture and session-aware chat interface.

AI-Powered Recruitment Platform

Streamlit, LangChain, Groq, Qdrant, HuggingFace, MongoDB, Llama LLMs, RAG

Developed an AI-powered recruitment platform that streamlines candidate screening using semantic resume-job matching with RAG technology. Built with Streamlit, Qdrant vector database, MongoDB, and Groq/Llama LLMs to automatically analyze resumes, generate job descriptions, and provide ranked job recommendations with similarity scores, transforming the hiring process for recruiters.

Vision Reasoning Surveillance System

PyTorch, OpenCV, YOLO, LangChain, Ollama, LLM, Streamlit, Telegram Bot

Developed a surveillance system to detect threats from real-time camera feed with the use of YOLO, and notify users with a detailed description of suspicious objects and scenes using LLM, LangChain and Telegram Bot. This project was done to gain experience with computer vision models, large language models and Intelligent systems.

Brain Tumour Detection Model

PyTorch, TensorFlow, OpenCV, U-Net, CNN, Alumentations, Kaggle

Developed a brain tumour detection model using the U-Net encoder-decoder architecture on MRI images. Implemented separate versions in PyTorch and Tensorflow. Used two popular 2D and 3D MRI image datasets from Kaggle. Used data augmentation methods using the alumentations library. Trained and evaluated on DICE, IoU and custom loss functions. Achieved ~90% accuracy.

Weekly Report Generator & Team Dashboard

FastAPI, MongoDB, React 19, TypeScript, Zustand 5, Tailwind CSS, LangChain, LangGraph, Groq, Docker

Engineered an asynchronous Weekly Report Generator and Team Dashboard with RBAC (Contributors, Managers, Admins), ISO calendar week management, immutable version snapshots, and live KPI analytics. Built an AI Assistant using LangGraph StateGraph, LangChain RAG, and Groq LLM with a 5-response rolling summarizer and session memory. Containerized with Docker Compose.

Data Analytics Dashboard

Python, SQL, Snowflake, ETL, XGBoost, Streamlit

Analysed LinkedIn job posting data using a real-time ETL pipeline using python, into Snowflake for storage and processing. Built regression and classification models using XGBoost to derive insights from job trends and requirements. Implemented real-time data feeds and visualized analytics through an interactive dashboard.

SKILLS

Languages: Python, TypeScript, JS, C#, Java, C, PHP, Dart

AI / ML: TensorFlow, PyTorch, Transformer, LangChain, LangGraph, ScikitLearn, CrewAI, XGBoost, Groq, Gemini, HuggingFace, OpenCV, Kafka, Ollama, LlamaIndex, Unsloth, VITS, n8n, Make, Retel.ai

Full Stack: FastAPI, React, Next.js, Node.js, Zustand, Tailwind CSS, Django, Streamlit, React Expo, Flutter, .NET, Codeigniter

Dev / ML / AIOps: Docker, GitHub Actions, WandB, Ansible, Terraform, Langfuse, Azure Bicep, Azure DevOps

Databases: MongoDB, PostgreSQL, MySQL, ChromaDB, Qdrant, Weaviate, Neo4j, PGvector, Supabase, MinIO S3

Cloud & Data: GCP, Azure, Snowflake, Streamlit Cloud, Google Vertex AI, Huggingface, Kaggle, Roboflow

Soft Skills: Critical thinking, Problem solving, Collaboration, Time management, Teamwork, Leadership, Adaptability

PUBLICATIONS & CONTRIBUTIONS

Musical Chord Recognition using Deep Learning

Research paper published in the journal at the NSRSIT'25 Conference.

accordo.ai

Published PYPI package with the developed DL Models for community use.

ACTIVITIES & EXTRACURRICULARS

- Bassist at In the Zone Band (2025 - Present), Unstoppable Band (2023 - 2025)
- Choir / Music Director at Music Club of NSBM (2023 - 2024)
- Member of the ICT Club, Maths & Science Society of St. Peter's College (2019 - 2021)
- Keyboardist / Choir Leader at Our Lady of Lourdes Choir
- Member of the College Under 13 Div. 1 & 3 Cricket team (2013 - 2014)
- President of the YCS Movement (2020 - 2021) & DH. Prefect of Our Lady of Lourdes Church (2017 - 2018)

REFERENCES

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